

CENG381 Stochastic Processes

Continuous-Time Markov Chains and Birth-Death Processes - Question Set 1

Notation used in this question set:

Q (rate matrix):	an $S \times S$ matrix where $Q(i,j) \geq 0$ for $i \neq j$ is the rate of jumping from state i to state j , and $Q(i,i) = -\sum_{j \neq i} Q(i,j)$. Every row of Q sums to 0.
$q_i = -Q(i,i)$:	the total rate of leaving state i . The chain spends an $\text{Exponential}(q_i)$ amount of time in state i before jumping.
Embedded chain P:	the discrete-time chain seen only at jump times. $P(i,j) = Q(i,j) / q_i$ for $j \neq i$ and $P(i,i) = 0$.
Global balance:	$\pi(i) * q_i = \sum_{j \neq i} \pi(j) * Q(j,i)$ for every state i . Rate of leaving state i = rate of entering state i .
Detailed balance:	$\pi(i) * Q(i,j) = \pi(j) * Q(j,i)$ for every pair i,j . This is a stronger condition -- it implies global balance, and if it holds the chain is called REVERSIBLE.

*Birth-Death processes are always reversible: $Q(i,j) = 0$ unless $|i-j| = 1$. Detailed balance gives: $\pi(n) = \pi(0) * (\lambda_0 * \lambda_1 * \dots * \lambda_{n-1}) / (\mu_1 * \mu_2 * \dots * \mu_n)$.*

Q1 (30 points). An air conditioner operates in states Off (0), Low (1), High (2). The rate matrix Q is:

	0	1	2
0	[-1/3	1/4	1/12]
1	[1/3	-1/2	1/6]
2	[1/6	1/3	-1/2]

- For each state i , identify q_i (the total leaving rate) from the diagonal of Q . Then write down the distribution of the holding time H_i (the time the chain spends in state i before jumping). Which state has the shortest expected holding time?
- Write the three global balance equations (one per state) using $\pi * Q = 0$, i.e., for each state i : $\pi(i) * q_i = \sum_{j \neq i} \pi(j) * Q(j,i)$. Do not solve yet -- just write them out explicitly.
- Solve the system from part (b) together with $\pi(0) + \pi(1) + \pi(2) = 1$ to find the stationary distribution π . Express each $\pi(i)$ as a fraction.

Q2 (35 points). A birth-death process on states $\{0, 1, 2, 3, \dots\}$ has birth rates $\lambda_n = \lambda$ (constant) and death rates $\mu_n = n * \mu$ (proportional to the current population). This models a system where each of the n individuals independently dies at rate μ .

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a) Write the rate matrix Q for states $\{0, 1, 2, 3\}$ (a 4×4 matrix). Leave entries in terms of λ and μ . Recall: from state n , births go to $n+1$ at rate λ_n and deaths go to $n-1$ at rate μ_n .

b) Apply detailed balance $\pi(n)\lambda_n = \pi(n+1)\mu_{n+1}$ repeatedly to derive the stationary distribution:

$$\pi(n) = \pi(0) * (\lambda/\mu)^n / n!$$

Then use $\sum_{n=0}^{\infty} \pi(n) = 1$ and the series $e^x = \sum x^n/n!$ to find $\pi(0) = e^{-(\lambda/\mu)}$. Write the final form of $\pi(n)$.

c) Using the stationary distribution found in (b), compute $E[N]$, the expected population size. Use $E[N] = \sum_{n=0}^{\infty} n * \pi(n)$ and the identity $\sum_{n=1}^{\infty} n * r^n / n! = r * e^r$ to evaluate the sum. What is $E[N]$ in terms of λ and μ ?

Q3 (35 points). A three-state CTMC on $\{A, B, C\}$ has rate matrix:

	A	B	C
A	[-5	3	2]
B	[2	-6	4]
C	[1	2	-3]

a) Check whether detailed balance holds between states A and B: compute $\pi(A)Q(A,B)$ and $\pi(B)Q(B,A)$ and determine whether they are equal for ANY stationary distribution π . If $Q(A,B)/Q(B,A) = \pi(B)/\pi(A)$ from detailed balance, what ratio $\pi(B)/\pi(A)$ would be required?

b) Write the global balance equations for this chain ($\pi * Q = 0$) as three linear equations. Use the equation for state A:

$$-5\pi(A) + 2\pi(B) + 1\pi(C) = 0$$

c) Solve the system from (b) together with $\pi(A) + \pi(B) + \pi(C) = 1$. Set up a 2×2 system by expressing $\pi(B)$ and $\pi(C)$ in terms of $\pi(A)$, then substitute into the normalisation equation.